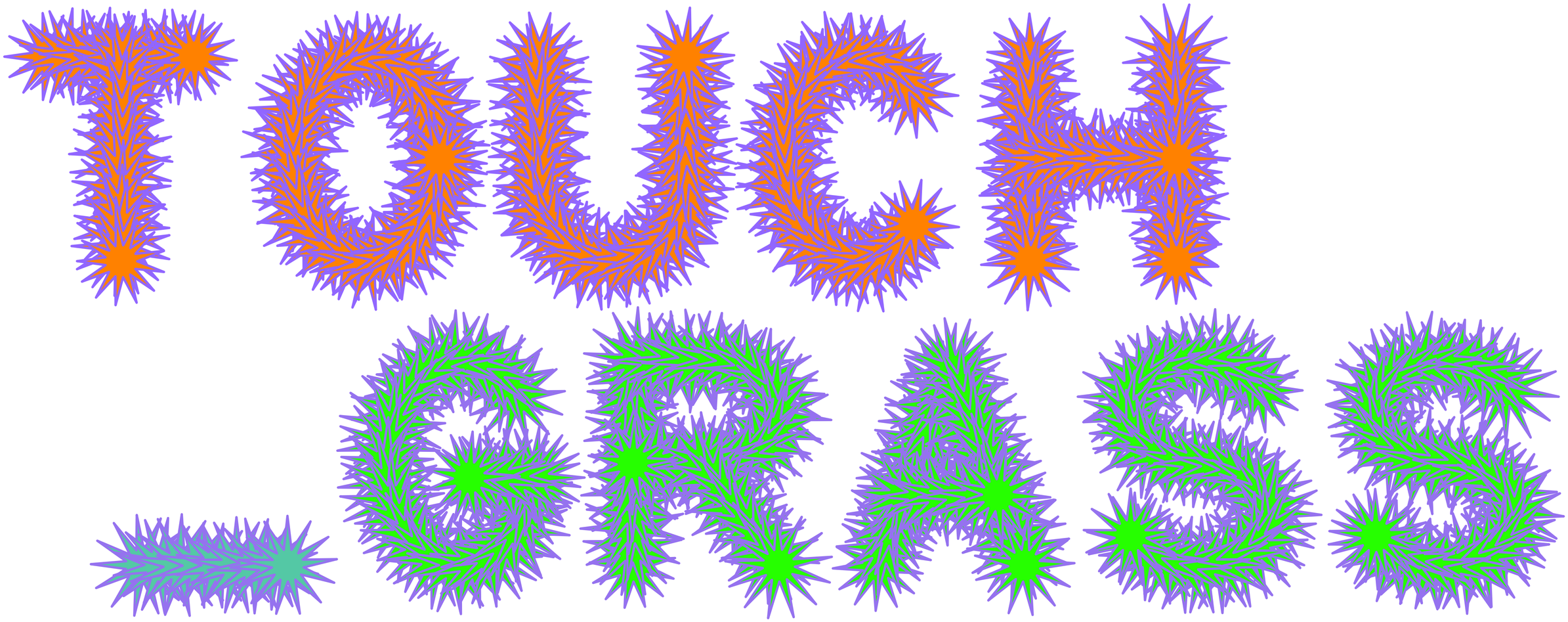


A workshop for disobedient websites that sense their environment, depend on it, and change with it.

Design, research, technology

Bureau 1217

When did a website last say no to you?

Everything. Always. On demand.

Click → instant answer

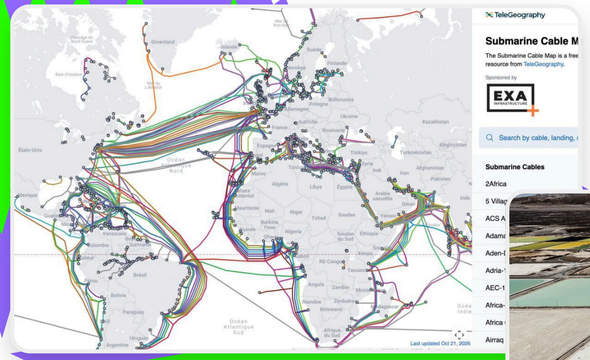
3:00 a.m. on a Sunday → open

Refresh → same page, again

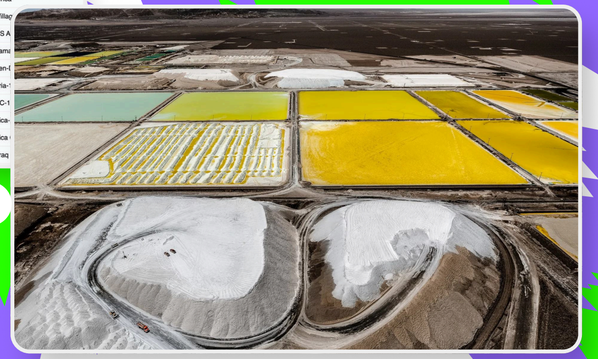
1 visitor or 1 million → same answer

24/7 · 365 · "99.9% uptime" → always on

↗ Timo Arnall, Internet Machine (2014)



↗ TeleGeography, Submarine Cable Map



↗ Tom Hegen, The Lithium Series

Behind every page, a world

↗ Trevor Paglen, Landing Sites

↗ Timo Arnall, Internet Machine (2014)

Take-off the leash

What if...

...a website only opened at low tide?

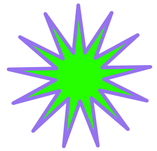
...it became tired when too many people visit it?

...it slept when the birds sleep?

...it waited until the room was quiet?

...it shook every time there's an earthquake on Earth?

One website that depends on something beyond the page.



What does it sense?

What does it depend on?

What changes?

Three design principles.

1

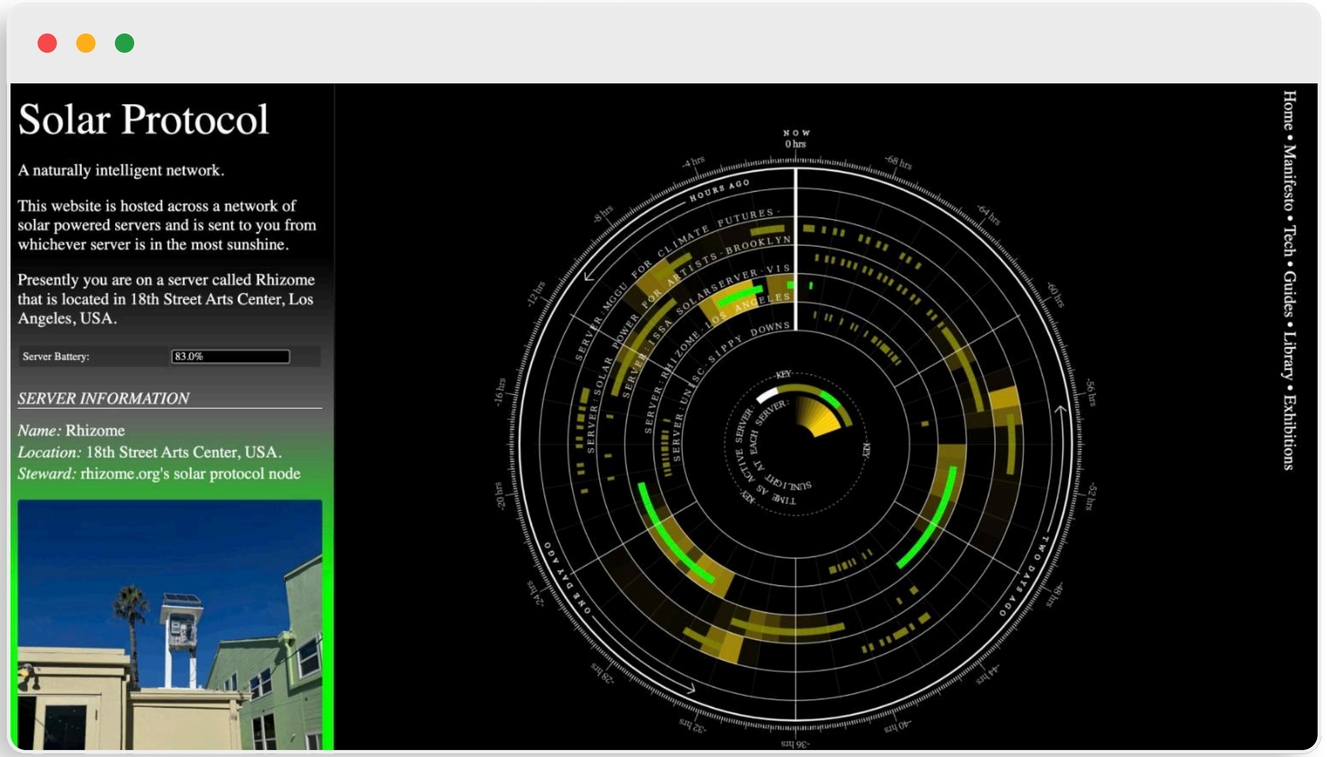
No constant availability.

Solar Protocol

A network of solar servers. The website comes from wherever the sun is shining.

SENSES	Sunlight, around the planet
DEPENDS ON	Solar energy
CHANGES	Which server answers you

↗ solarprotocol.net



Chris Bolin, Offline Only

An article you can only read once you disconnect from the internet. Being online is the obstacle.

SENSES	Your connection status
DEPENDS ON	Being offline
CHANGES	Whether the text appears

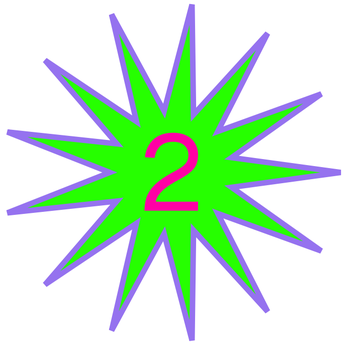
↗ chrisbolin.co/offline/

Aram Bartholl, Keepalive (2015)

A boulder with a router inside. Build a fire next to it: a thermoelectric generator powers an offline library of survival guides.

SENSES	Heat
DEPENDS ON	Your fire
CHANGES	Whether the network is on

↗ arambartholl.com/keepalive/



At least one condition comes

from the environment.

Nikki Makagiansar & Todd Whitney, Unearthing Air

An article about indoor air. The more particles in the air, the more letters disappear: you can read less. Hover a hidden letter to see what caused the spike, cooking, cleaning, a candle.

SENSES	Particulate levels over 24 hours
DEPENDS ON	Everyday life in a home
CHANGES	How much of the text you can read

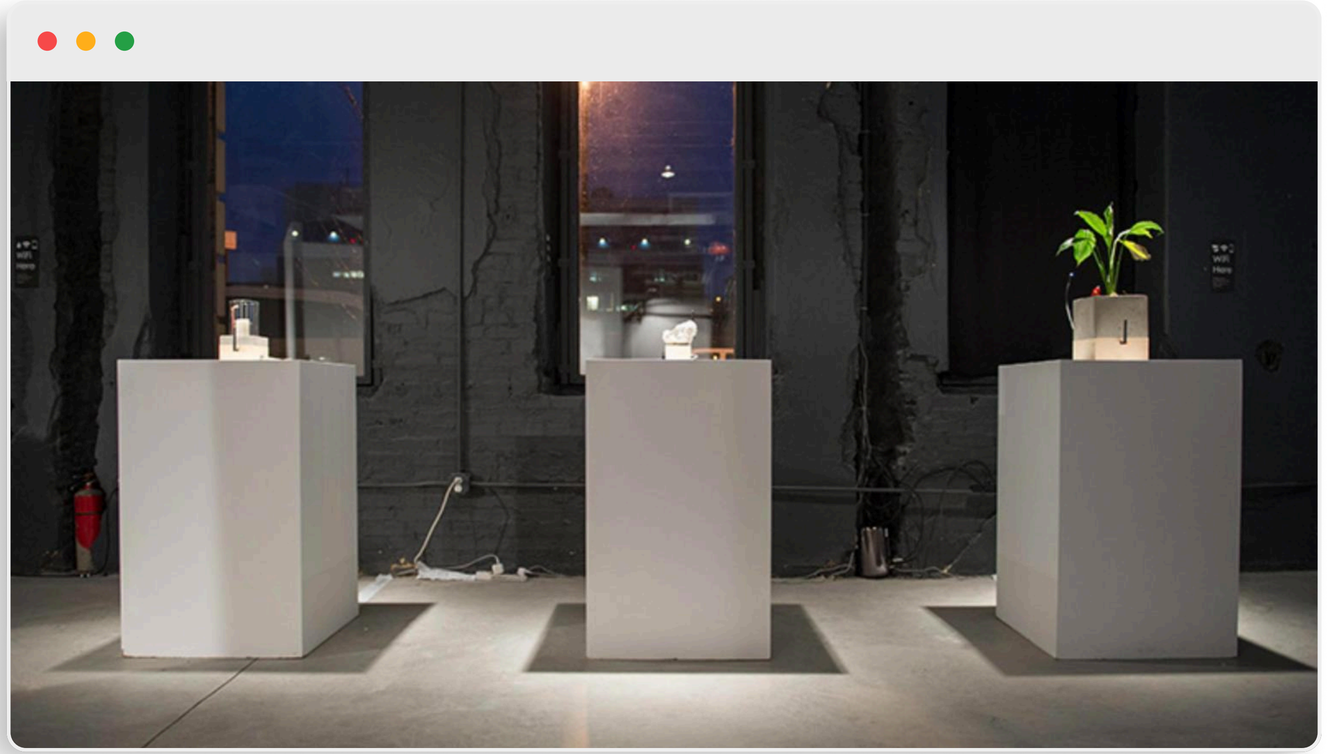
↗ niktari.com/projects/unearthing-air

Tega Brain, Being Radiotropic (2016)

Wifi routers that negotiate with non-human systems: one follows the moon's phase, one exists only while a candle burns, one is controlled by a houseplant.

SENSES	The moon, a flame, a plant's posture
DEPENDS ON	Natural and living cycles
CHANGES	Whether the network exists

↗ tegabrain.com/Being-Radiotropic

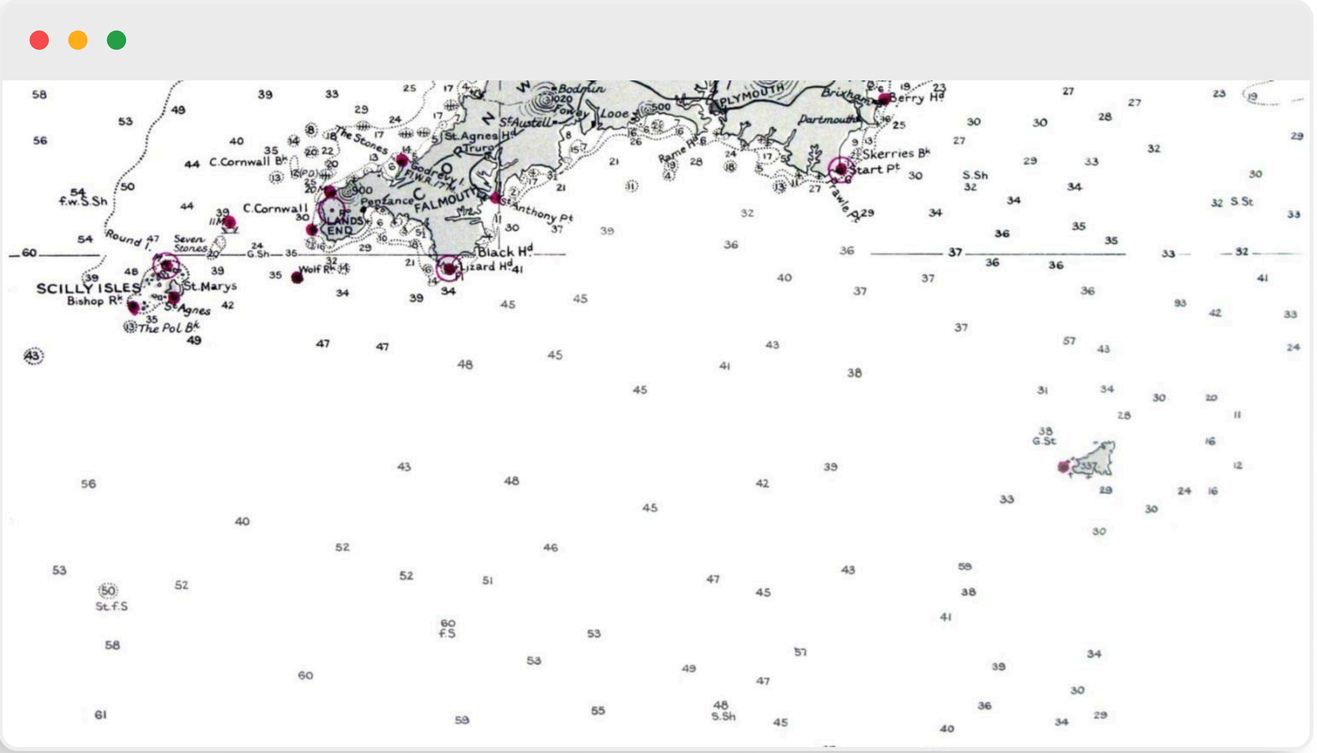


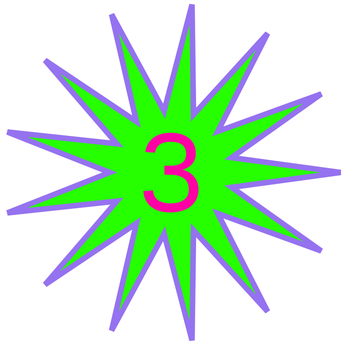
J. R. Carpenter, This is a Picture of Wind

A poetic almanac and live wind report for South West England: today's wind speed and its Beaufort description shape the text you read.

SENSES	Live wind speed
DEPENDS ON	The weather in South West England
CHANGES	The words on the page

↗ luckysoap.com/apictureofwind/main.html





The condition gives it a behaviour,

like something alive.

Raphaël Bastide, Snow on GPU (2021)

Asterisks fall like snow down the page and melt as they reach the place where the laptop's GPU sits. The page makes the computer work: its own heat melts the snow.

SENSES	The position of the GPU in the laptop
DEPENDS ON	The computer's own effort
CHANGES	Snow melts where the machine heats

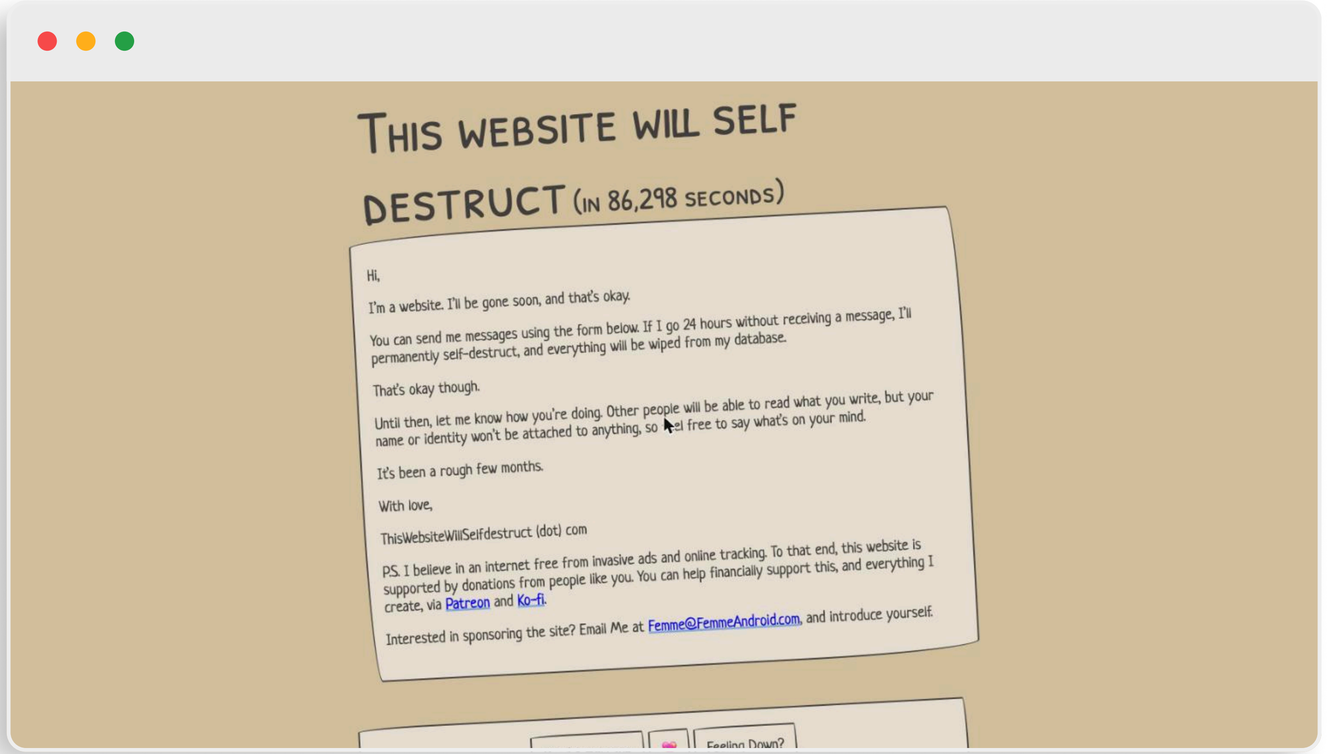
↗ snow-on-gpu.oo8.be/

This Website Will Self-Destruct

If nobody writes to it for 24 hours, it deletes itself. In December 2023, nobody wrote.

SENSES	Messages from visitors
DEPENDS ON	Our attention
CHANGES	It exists, or disappears forever

↗ thiswebsitewillselfdestruct.com

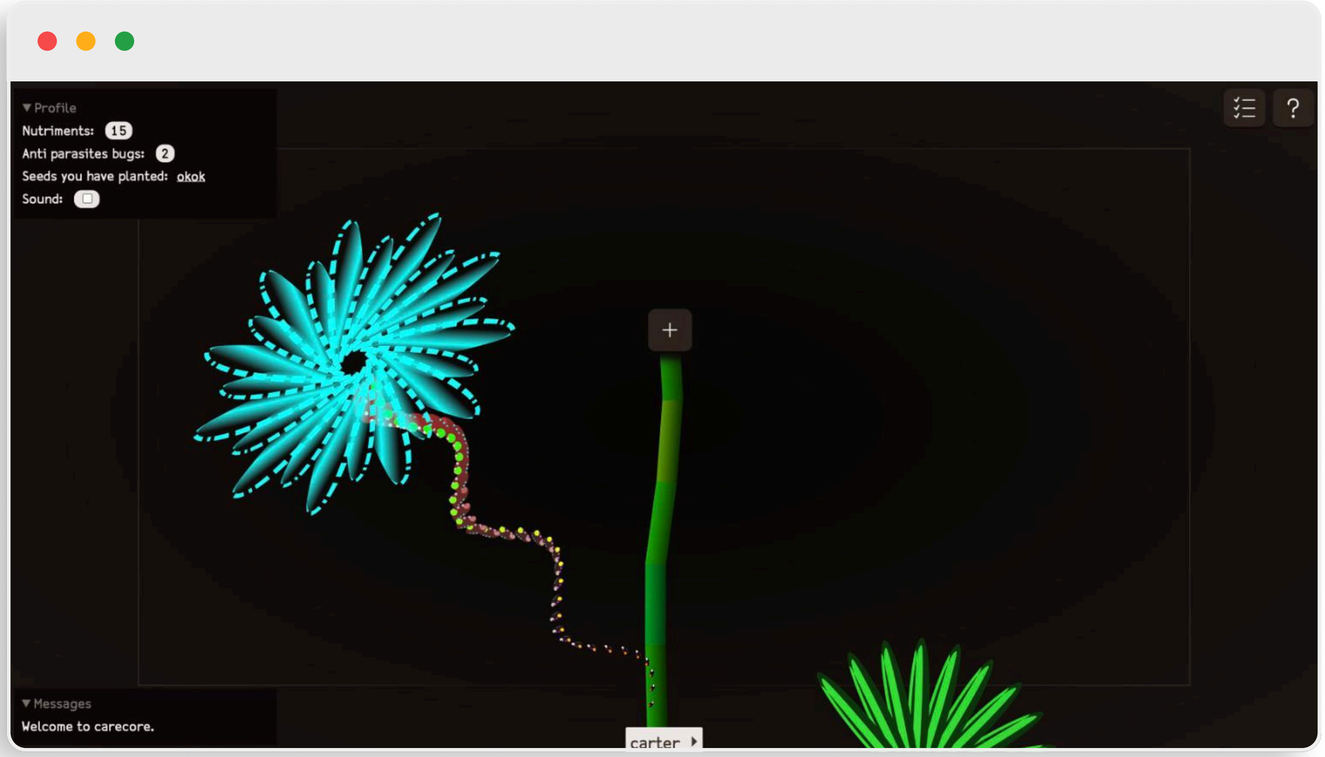


Raphaël Bastide, CareCore

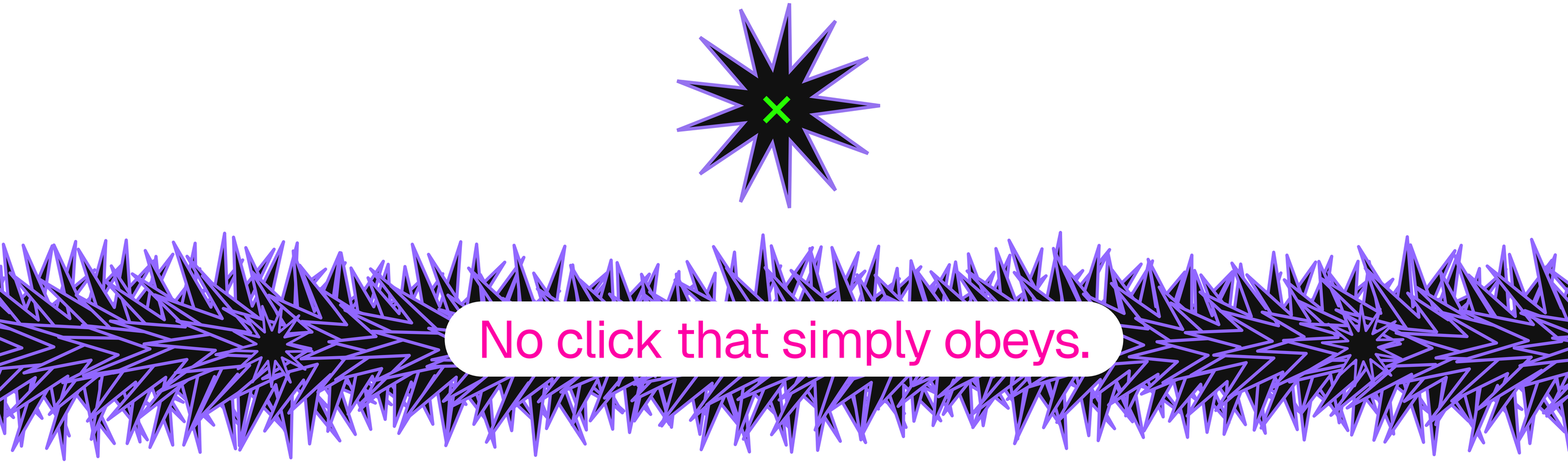
A garden shared by everyone who visits. Plant one seed a day, feed it, treat it against parasites, anyone can care for anyone's plant. Every plant dies at some point; compost it to feed the others.

SENSES	Visitors' care: feeding, treating, composting
DEPENDS ON	Others coming back, day after day
CHANGES	What grows, blooms and dies

↗ raphaelbastide.com/carecore/#p185



Three constraints against the servile web.
The mouse is allowed. Obedience is not.

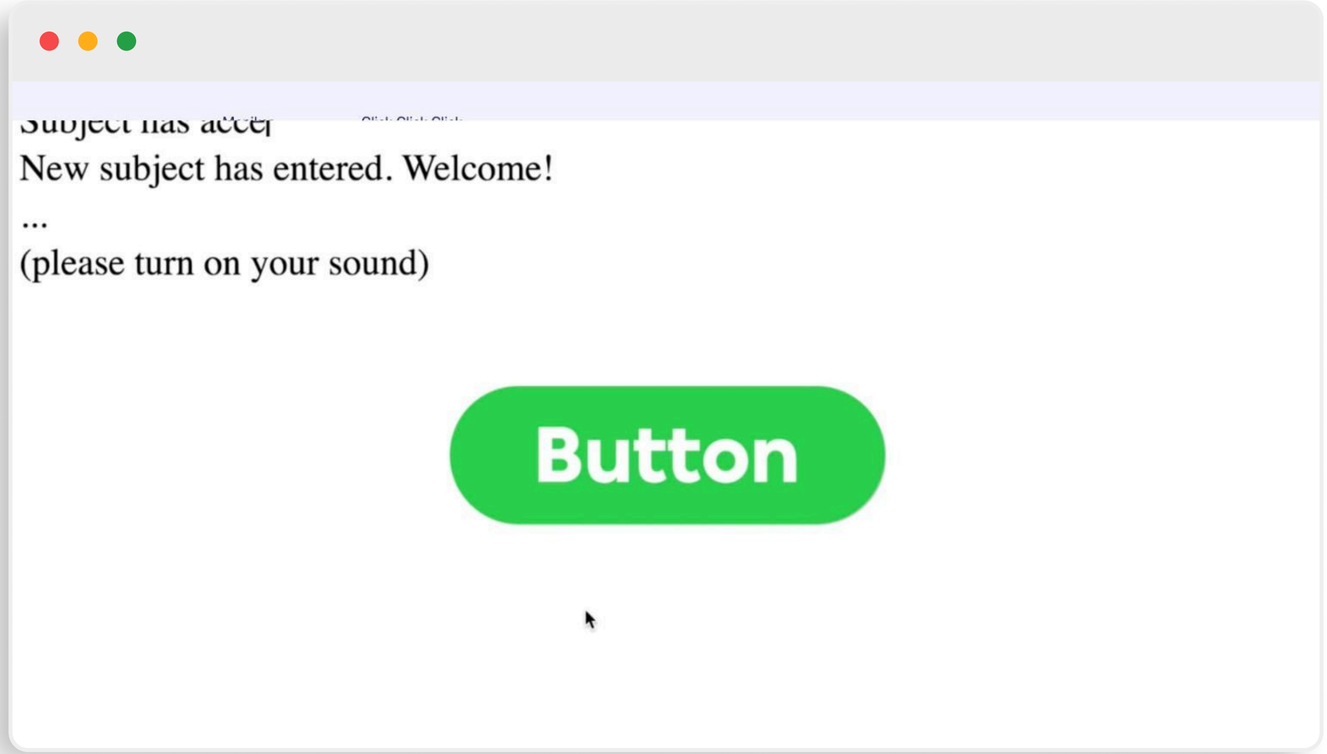

No click that simply obeys.

Moniker × VPRO: Click Click Click (2016)

One green button. Every move you make is measured, described and judged, then compared with everyone before you.

SENSES	Every mouse event
DEPENDS ON	Previous visitors' behaviour
CHANGES	The verdict on you

↗ studiomoniker.com/projects/click-click-click

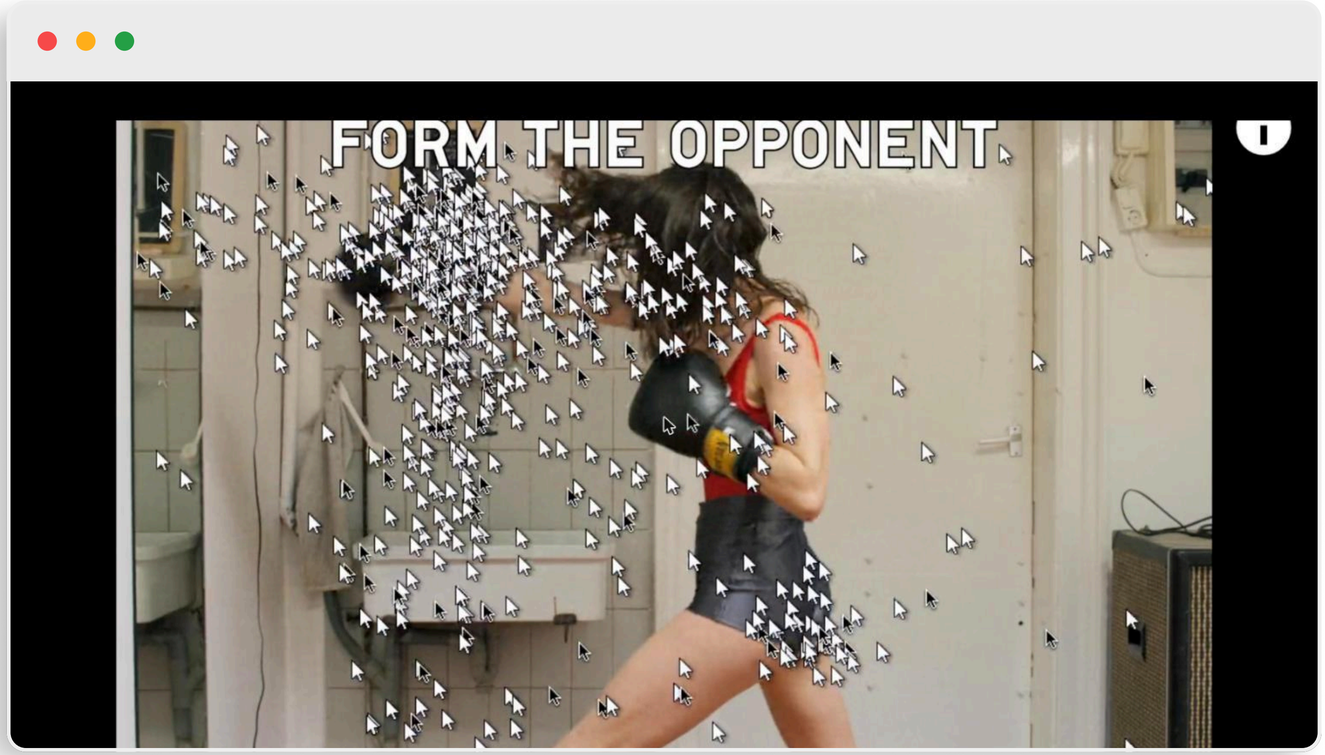


Moniker, Do Not Touch (2013)

A music video for Light Light's "Kilo". You follow instructions with your cursor; the cursors of everyone before you appear in the video, re-rendered every hour.

SENSES	Your cursor
DEPENDS ON	Everyone who came before you
CHANGES	The video itself

↗ donottouch.org/

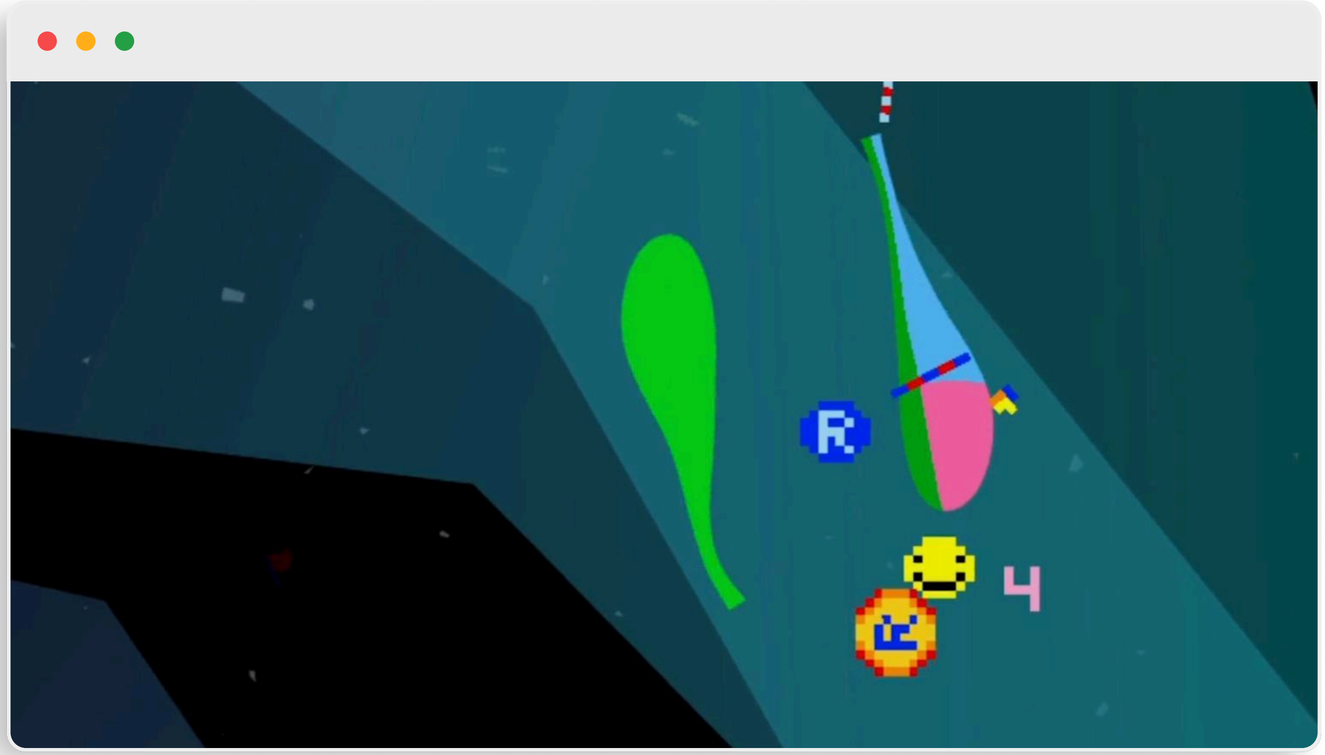


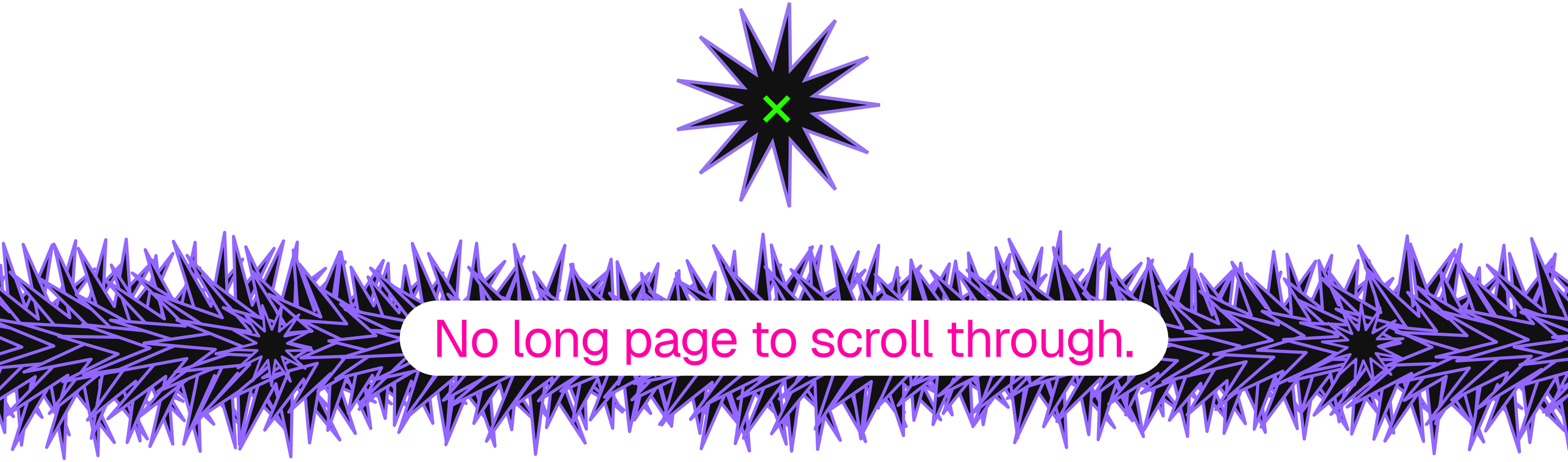
Moniker for M+, Touch for Luck

Keep your finger on your phone without letting go, together with others, in a shared pond projected on the M+ façade in Hong Kong.

SENSES	A touch held over time
DEPENDS ON	Others touching at the same moment
CHANGES	Your fish in the pond

↗ studiomoniker.com/projects/touch-for-luck





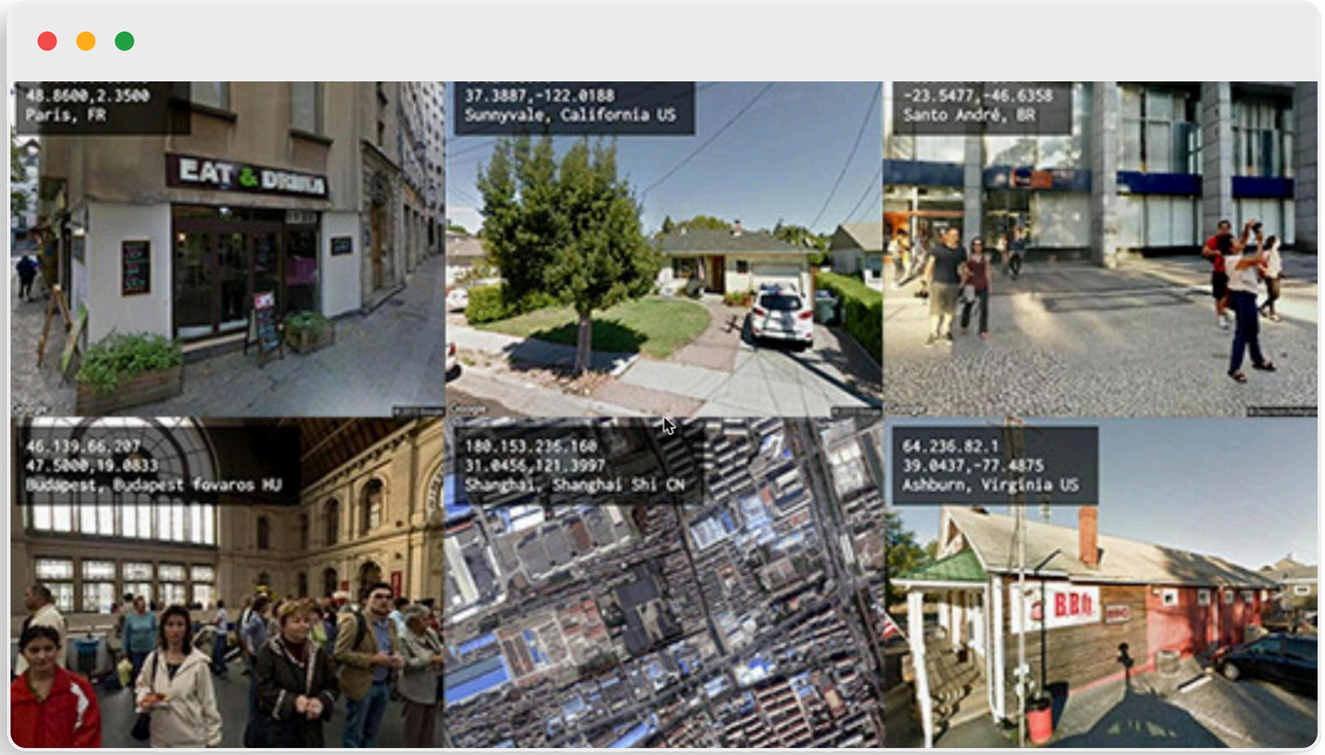
No long page to scroll through.

Ben Grosser, Tracing You (2015)

From your IP address alone, the site traces your connection back to a possible origin and shows the closest image it can find of where you might be.

SENSES	Your IP address
DEPENDS ON	Online data about places
CHANGES	A picture of "your" surroundings, right, or far off

↗ tracingyou.bengrosser.com/

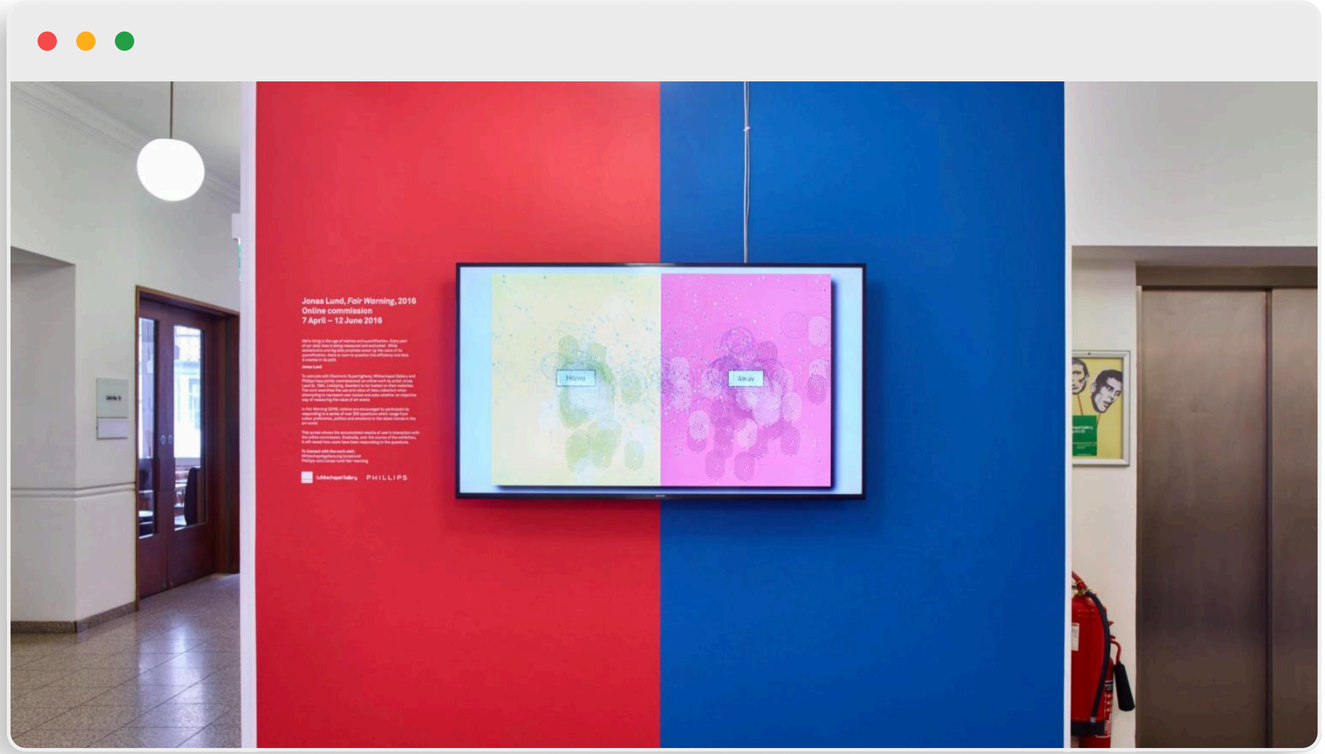


Jonas Lund, Fair Warning (2016)

Over 300 questions: click an answer, or wait and the next one comes. Everyone's clicks and cursor movements are shown together.

SENSES	Clicks, cursor, hesitation
DEPENDS ON	Other visitors
CHANGES	The shared picture of answers

↗ jonaslund.com/works/fair-warning/



Moniker for Mozilla, Paperstorm.it (2017)

Float above a landmark and click to drop a leaflet. Everyone's drops pile up on the building into one shared protest.

SENSES	Every visitor's clicks
DEPENDS ON	Others dropping at the same time
CHANGES	The pile of leaflets on the building

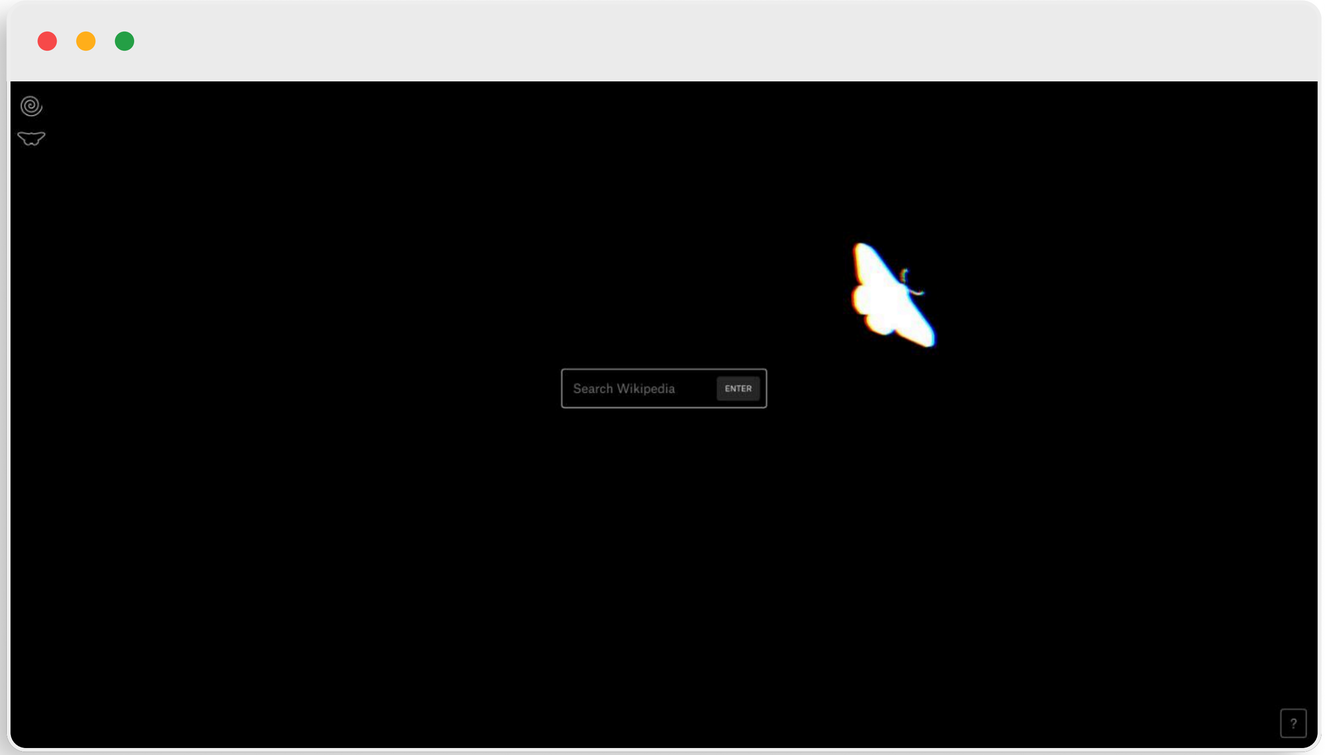
↗ studiomoniker.com/projects/paperstorm

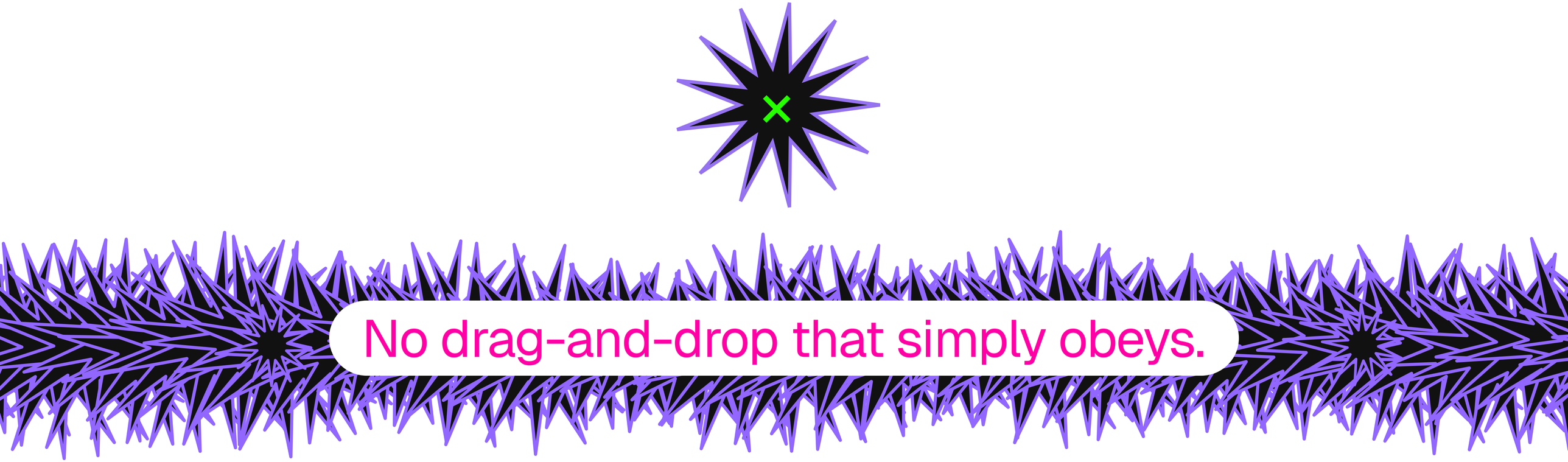
Shaheer Tarar, (moth) in search

A search engine run by a moth: it crawls Wikipedia and is drawn to hyperlinks like light. An offshoot of (moth) in relay, a project about internet infrastructure.

SENSES	Hyperlinks, as sources of light
DEPENDS ON	The live structure of Wikipedia
CHANGES	Where the search wanders

↗ moth-in-relay.net/search/



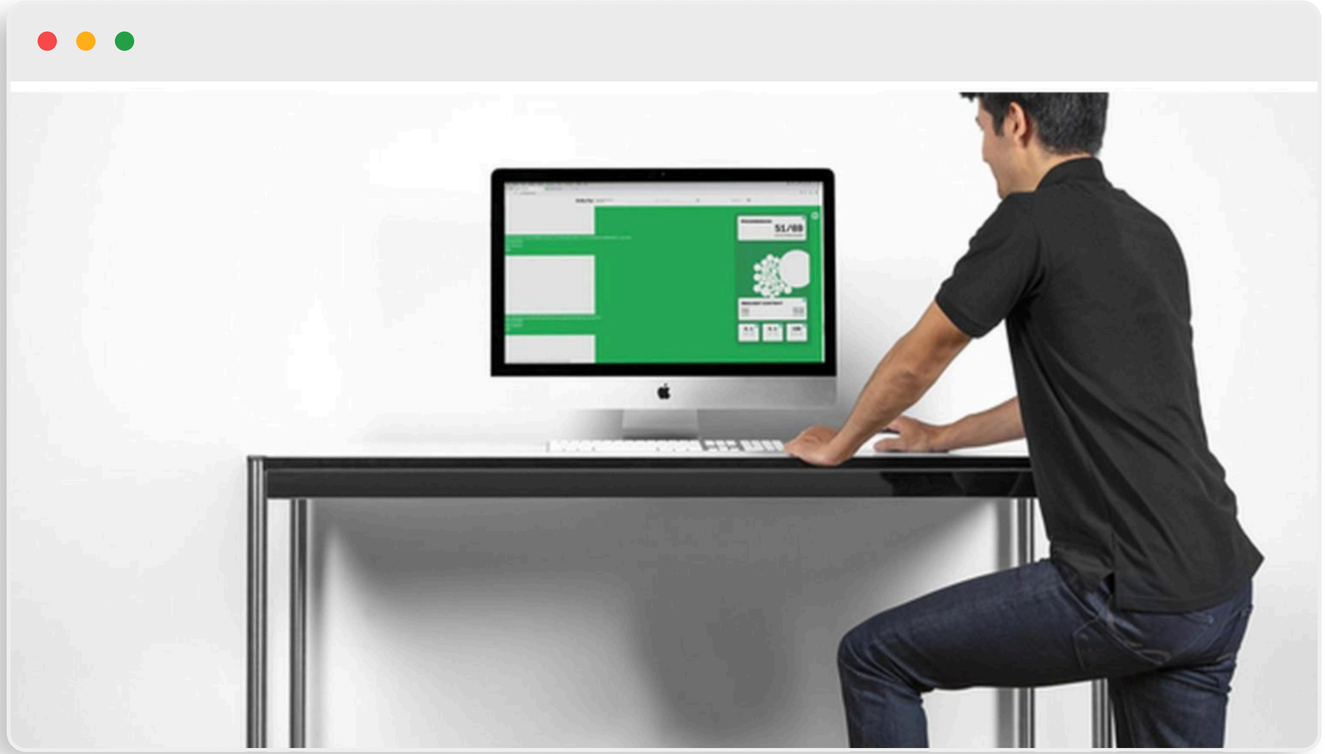

No drag-and-drop that simply obeys.

Aurélien Pellegrini, Pump and Surf (ECAL, 2020)

Before a website appears, you make a physical effort roughly equal to the energy needed to transmit its data.

SENSES	Your physical effort
DEPENDS ON	The weight of the website's data
CHANGES	Access to the page

↗ ecal.ch/en/feed/projects/6766/pump-and-surf/

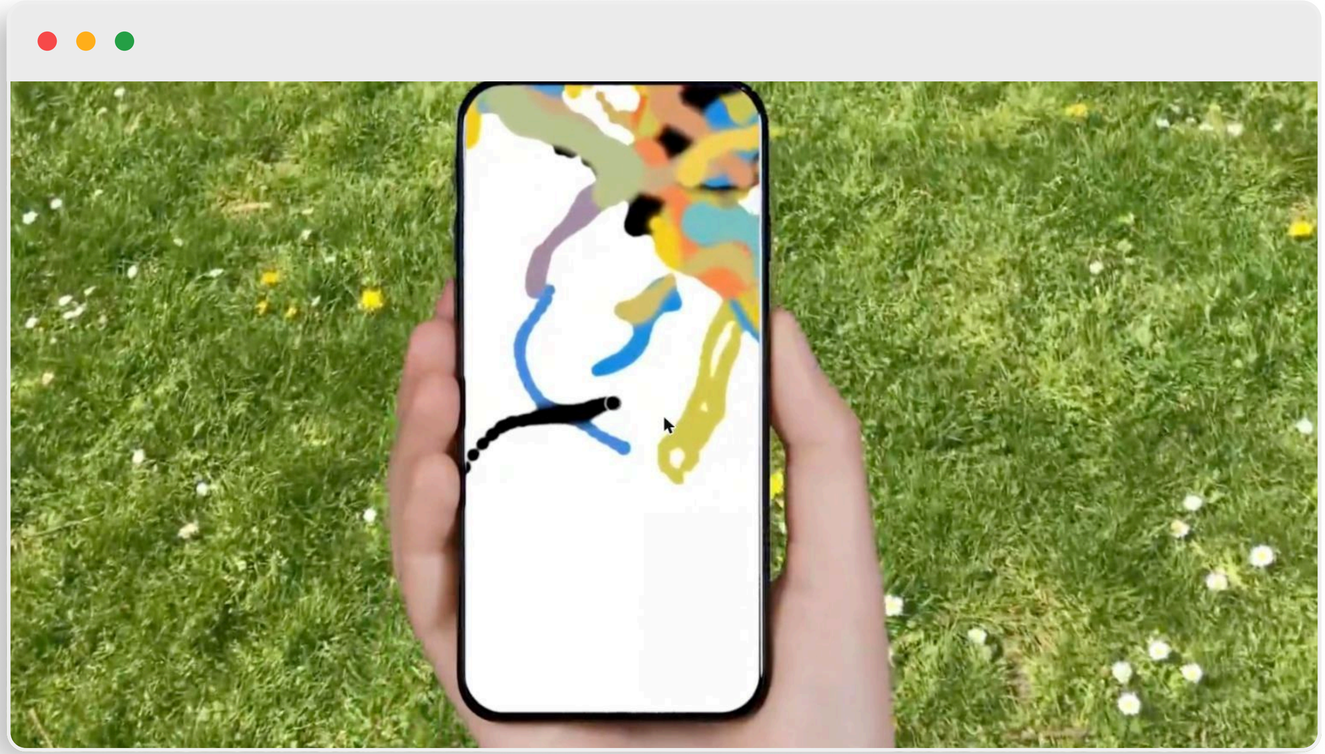


Moniker, Painted Earth (2018)

Your phone is a brush, the planet your canvas. Walk fast and lines turn into drops; stand still and the ink pools. Paintings stay where they were made.

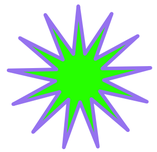
SENSES	Your position (Galileo satellites) and speed
DEPENDS ON	Being physically there
CHANGES	A shared painting on the map

↗ studiomoniker.com/projects/painted-earth



Putting it all together.

From remote control to relationship.



SERVILE INTERACTION: the remote control		DISOBEDIENT INTERACTION: a relationship
Point, click, instant answer.	→	A click that leaves a trace, or needs someone else. ↗ Click Click Click · Do Not Touch · Paperstorm.it · (moth) in search
Scroll, and get more.	→	Content that arrives with time, weather or light. ↗ This is a Picture of Wind · Unearthing Air
Drag and drop: it obeys.	→	Things with their own weight: they resist, drift, come back. ↗ Painted Earth · Snow on GPU
Always on, the same for everyone.	→	It opens, sleeps or changes with conditions, and with who is there. ↗ Solar Protocol · Touch for Luck · CareCore
A finger giving orders.	→	A body that is present: effort, place, patience. ↗ Pump and Surf · Keepalive · Dead Drops

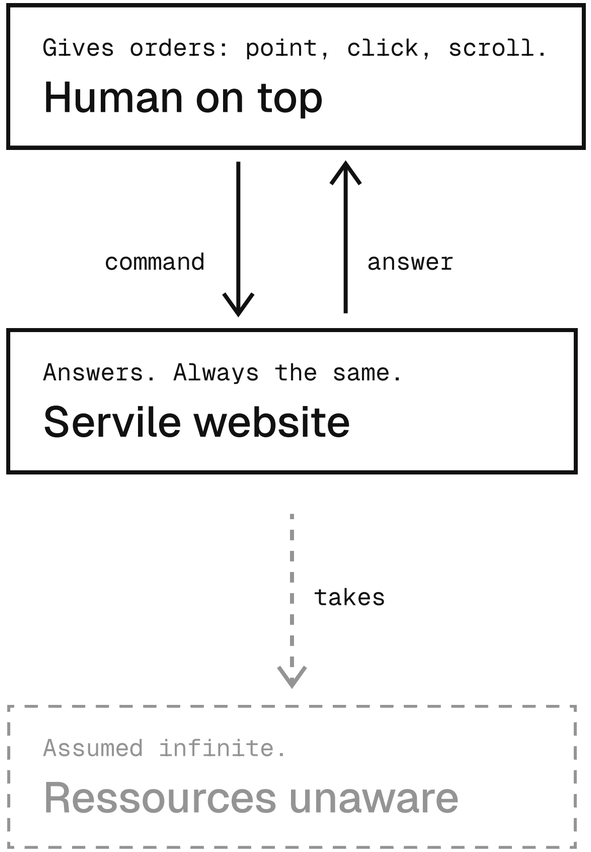
Chain of command vs. ecosystem.

A SERVILE WEBSITE

a chain of command, top-down, one direction

HOW WE COMMAND IT

point and click
scroll for more
drag and drop
refresh, search, swipe



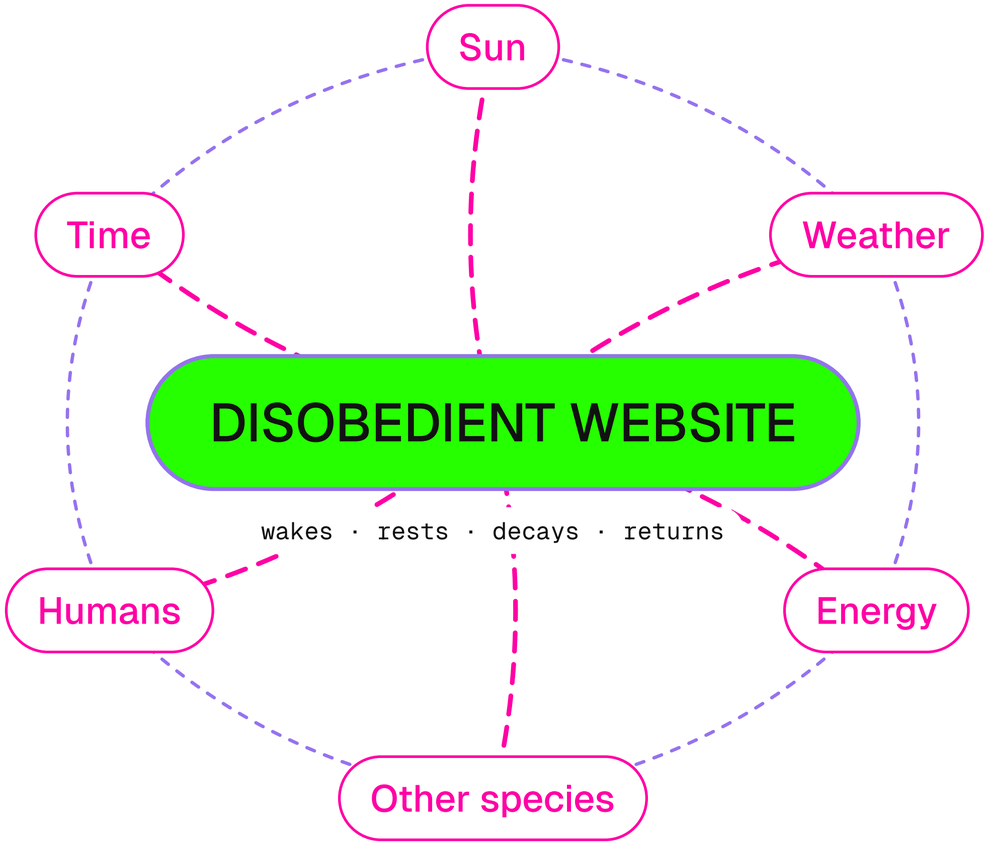
HOW IT ANSWERS

instantly
identically
for everyone
always on

Chain of command vs. ecosystem.

STEP 1 · GO OUTSIDE: the website depends on the world

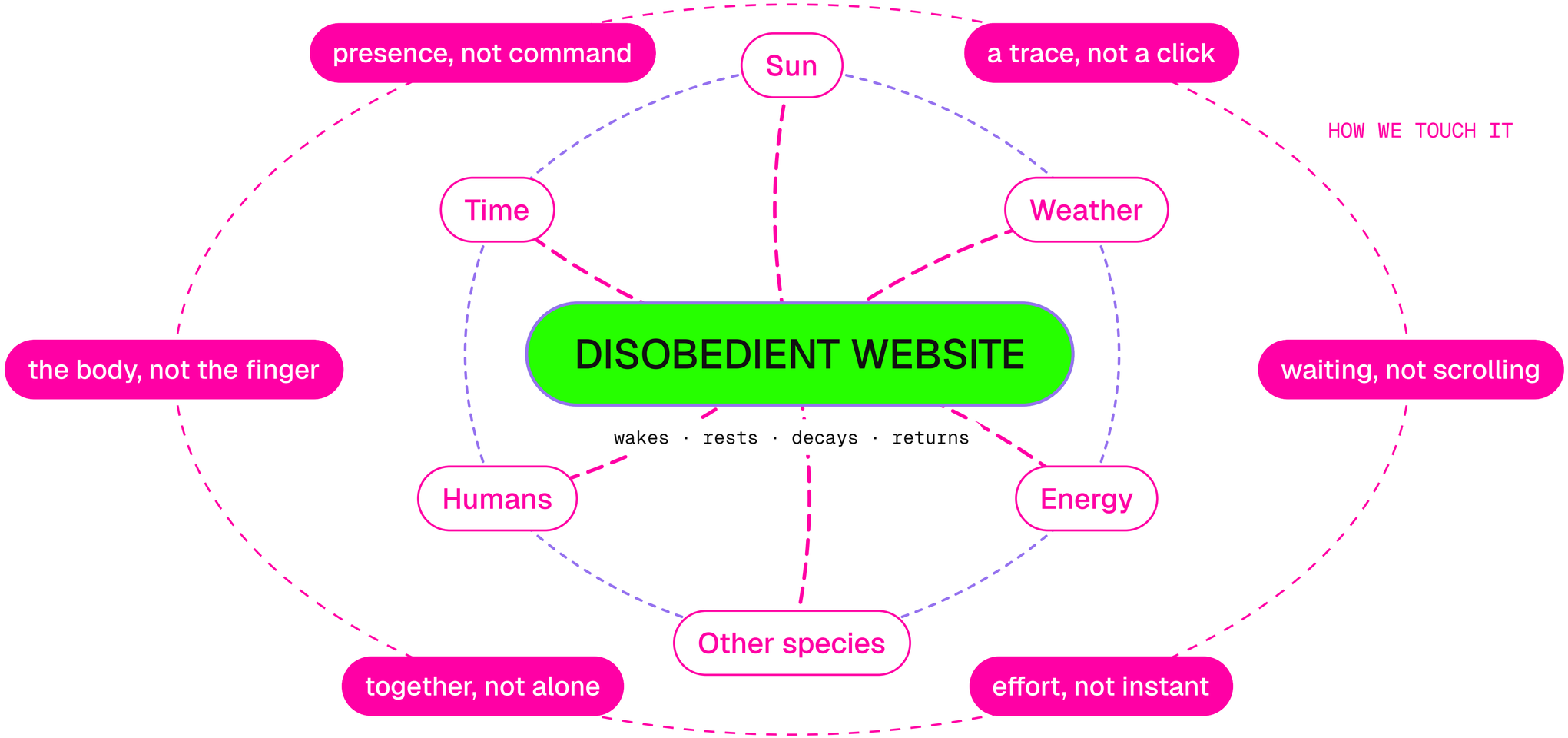
A TOUCH_GRASS WEBSITE



Chain of command vs. ecosystem.

STEP 2 · CHANGE THE RELATIONSHIP: how humans interact with it

A TOUCH_GRASS WEBSITE



What we expect

01

SIGNAL

What it senses.

Where does it come from?
How often does it change?
What happens when it's missing?

→

02

RULE

What it does with it.

What exactly triggers a change?
Threshold, duration, memory?
Can you say it in one sentence?

→

03

BEHAVIOUR

How it lives.

Which states can it be in?
Can it get tired, rest, recover?
How do we feel what it consumes?

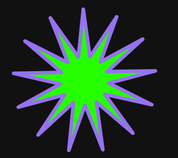
→

04

STORY

Why did it touch grass?

What does it change for you?
Can we feel what it depends on?
What stays out of our control?



- × ~~A vibecoded website you can't explain.~~
- × ~~A normal website with a weather widget.~~
- × ~~Live data as decoration.~~
- × ~~A background that turns blue when it rains.~~
- × ~~A demo of a sensor or an API.~~
- × ~~A website that works the same without its condition.~~

If you remove the condition and it still works, it's decoration.

26.10 TODAY

MON

Explore & formulate

· Intro, groups, rules
· Find your relationship with the world
· No code. No AI.

→ a concept + a first scenario

27.10 AI ON

TUE

Define & start building

· 9:00, present your concept
· Feedback → validation
· Setup: GitHub + Node.js
· Vibecoding starts

→ a validated system + a first interaction

28.10

WED

Prototype

· Build, test, break
· Technical & concept support

→ signal → rule → behaviour working

29.10

THU

Prototype & narrate

· Give it states and a story
· Test with other groups
· Test in real conditions

→ a living prototype + its scenario

30.10 RELEASE

FRI

Test, document, release

· Document on GitHub
· Final presentations
· Let the conditions interfere

→ live website + README

The final → a live website presentation + a GitHub repo.

- What did you observe?
- Which condition / API do you use?
- How does your website behave?
- What are its limits?
- + a short live demo

Your toolbox for the week.

↗ [workshop URL]

Brief & principles

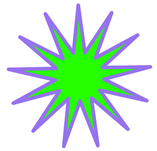
Schedule, day by day

4 experiments, when you're stuck

Resources, Read · Look · Connect · Build

Let's make the web disobedient.

One website, many disobedient websites.



↗ touch-grass.bureau1217.ch

Every project is published on the workshop website.	01
Your website can listen to another group's website.	02
It can react to the workshop site, or change it.	03
Together, they become one ecosystem.	04

Tomorrow, 9:00: your concept in 5 slides.

01

THE IDEA

Your website in one sentence.

What it senses, what it depends on, what changes.

02

STEPS

How it unfolds.

First visit, coming back, when the condition is missing.

03

INTERACTION

How we touch it.

Which clues tell the visitor how to interact? No remote control.

04

RESOURCES

What it consumes.

Energy, data, attention. How do we feel it in the behaviour?

05

DATA

What it listens to.

Which API, sensor or device? How often? What if it's missing?

+ QUESTIONS & ANSWERS

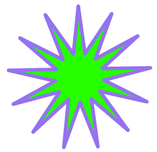
Interaction ideas.

INSTEAD OF…	TRY…
Clicking a button.	Hold it, wait for it, or need two people at once. ↗ pointerdown / pointerup · setTimeout · shared state
Scrolling a feed.	Let time, light or weather reveal the content. ↗ Date · SunCalc · Open-Meteo
Typing in a search box.	Speak, whisper, or stay silent. ↗ getUserMedia · Web Audio (volume)
Moving the cursor anywhere.	Your speed, hesitation or stillness changes things. ↗ pointermove · speed · idle time
Opening it whenever you want.	Come back at the right moment: dawn, wind, full moon. ↗ SunCalc · Open-Meteo · localStorage (memory)

Signals you can listen to.

API · WHAT IT GIVES	IDEA · WHAT IT COULD CHANGE
Open-Meteo · wind, rain, temperature, UV	→ The page leans when the bise blows. ↗ open-meteo.com · no key
Sunrise-Sunset / SunCalc · sun position, dawn, dusk	→ The website only opens at golden hour. ↗ sunrise-sunset.org · suncalc (npm) · no key
USGS Earthquakes · live earthquakes worldwide	→ It shakes every time the Earth does. ↗ earthquake.usgs.gov · GeoJSON feed · no key
existenz.ch · Rhône flow, lake temperature	→ The text flows faster when the river is high. ↗ api.existenz.ch · Swiss hydrological data
eBird · recent bird observations nearby	→ It sleeps when the birds sleep. ↗ ebird.org · API · free key

Things you can plug in.



DEVICE · WHAT IT SENSES		IDEA · WHAT IT COULD CHANGE
Smartphone · tilt, shake, compass, mic, camera	→	**Tilt the phone like a leaf in the wind. Whisper to it.** ↗ DeviceOrientation · DeviceMotion · getUserMedia · HTTPS + permission on iOS
MIDI controller · knobs, faders, pads	→	**Slow knobs instead of clicks: a fader for patience.** ↗ Web MIDI API · Chrome, Edge, Firefox
Webcam & microphone · light, presence, noise	→	**The page dims with the room and waits for silence.** ↗ getUserMedia · canvas brightness · Web Audio
Arduino / micro:bit · light, humidity, soil, CO_2	→	**A plant in the room becomes the signal.** ↗ Web Serial API (Chrome) · or a small Node.js bridge
Your laptop itself · battery, network, CPU load	→	**It gets tired when the battery is low.** ↗ Battery Status API (Chromium) · Network Information · Compute Pressure

Aram Bartholl (curator), Offline Art: new2 (2013)

Net art broadcast from wifi routers with no internet. Each work has its own network: to see the next one, you walk to the next router.

SENSES	Your presence in the room
DEPENDS ON	Wifi range
CHANGES	Which artwork you can open

↗ arambartholl.com/offline-art-new2/